

DECISION SCIENCES INSTITUTE

From Pixels to Insights: Large Language Models
for Automated Data Visualization Enhancement

Dr. Andres Fortino
New York University
agf249@nyu.edu

ABSTRACT

This study explores the potential of Large Language Models (LLMs) as visualization coaches using a structured analytical framework. We trained ChatGPT with an 18-point visualization assessment framework and evaluated its ability to analyze and suggest improvements for data visualizations from a White House economic policy presentation. The research compared ChatGPT's analysis against a Subject Matter Expert (SME) across five diverse chart types, with agreement levels ranging from 81.4% to 93.3%. The highest concordance appeared in functional aspects of visualization (>90%), while the greatest divergence emerged in technical recommendations and chart type selection (81.4%). ChatGPT successfully generated improved versions of the charts, focusing on enhanced clarity and accessibility for general audiences. While the results demonstrate the feasibility of using AI as a visualization coach, they also highlight important limitations in technical depth and context awareness. This proof-of-concept study suggests that LLMs can effectively complement human expertise in data visualization, particularly for basic analysis and accessibility improvements, while indicating areas needing further development for more sophisticated applications.

Keywords: Data visualization, Large language models, ChatGPT, Visual analytics, Human-AI collaboration, Decision support

INTRODUCTION

Our study aims to develop a chatbot that analyzes and improves data visualizations. This tool will assist data analysts by automatically evaluating and suggesting enhancements to their charts, making them more effective for communication and decision-making. This project addresses the growing need for high-quality data visualizations in businesses to ensure that insights are accurately and effectively conveyed.

The Problem

In the current data-driven business environment, there is a growing need for high-quality data visualizations to communicate insights and support decision-making effectively. However, creating clear, accurate, and visually appealing data visualizations often requires specialized skills and expertise that not all data analysts possess. Traditional methods of improving data visualizations can be time-consuming and subjective, leading to inconsistent quality across different analysts and organizations. There is a need for a more efficient, standardized, and accessible approach to enhancing data visualizations that can assist data analysts in improving their charts consistently and effectively.

Traditionally, applying such a checklist would require significant time and expertise from a human analyst. However, the chapter also explores an exciting new frontier: the use of large language models (LLMs) in data visualization analysis. By fine-tuning an LLM with the principles

embodied in the 18-point checklist, we create a powerful AI assistant capable of rapidly analyzing and suggesting improvements for data visualizations.

The paper concludes by comparing the manual analysis process with the AI-assisted approach. This comparison highlights both the progress made in automating complex analytical tasks and the continued importance of human judgment and creativity in data visualization. While the AI can quickly identify potential issues and suggest improvements based on established principles, the human analyst brings contextual understanding, creative problem-solving, and the ability to make nuanced decisions that go beyond rule-based assessments.

By presenting a comparison of both traditional and AI-assisted methods, this paper equips readers with a versatile toolkit for creating and improving data visualizations. Whether working through the list manually to develop a deep understanding of visualization principles or leveraging AI to streamline the analysis process, readers will gain valuable skills for communicating data effectively in various professional contexts.

Theory

The project is based on the concept that a chatbot utilizing Large Language Models (LLMs) and established data visualization principles can effectively analyze and offer valuable suggestions for improving data visualizations. This approach relies on the idea that AI can be trained to recognize and apply best practices in data visualization, as developed by Professor Fortino, to identify areas where charts may lack clarity, accuracy, and visual appeal. By automating this analysis and recommendation process, the chatbot could potentially provide consistent, objective, and actionable feedback for enhancing data visualizations, thereby improving the communication of insights across various professional settings.

Research Questions

1. How can an AI-driven chatbot effectively analyze and suggest improvements for data visualizations based on established principles?
2. To what extent can a chatbot provide valuable and actionable feedback for enhancing the clarity, accuracy, and visual appeal of data visualizations compared to traditional methods?
3. What is the potential impact of an AI-based visualization enhancement tool on the quality and consistency of data presentations in professional settings?

Hypothesis

An AI-driven chatbot, leveraging established data visualization principles, can provide valuable and actionable suggestions for improving data visualizations, resulting in measurable enhancements in clarity, accuracy, and visual appeal compared to the original charts, as demonstrated through expert evaluations and comparative analysis of visualizations before and after chatbot-suggested improvements.

LITERATURE REVIEW

The framework proposed by Fortino (2022) for analyzing and improving visuals consists of six dimensions: Story, Signs, Purpose, Perception, Method, and Charts. These dimensions provide a holistic approach to evaluating and enhancing data visuals. Each dimension focuses on specific aspects of effective visualization and is rooted in the work of known scholars in the discipline, as annotated below:

1. Story: Ensuring the visual tells a clear narrative
2. Signs: Using appropriate symbols and maintaining a high signal-to-noise ratio

3. Purpose: Aligning with organizational needs and audience expectations
4. Perception: Leveraging principles of visual perception for clarity
5. Method: Employing best practices in color use, decluttering, and titling
6. Charts: Selecting and implementing the most appropriate chart types

Within these dimensions, an 18-element checklist (three in each dimension) serves as a practical tool for assessing and refining data visualizations. It prompts creators to consider crucial elements such as the clarity of the visual story, the effectiveness of signs and symbols, the match between the visualization and its intended purpose, the application of perceptual principles, the judicious use of visual methods, and the selection of appropriate chart types.

Story

Al Shallowly (2011), an expert visual storyteller, tells us that: “Visualizations act as a campfire around which we gather to tell stories.” *Are we telling compelling business stories with our visuals?*

Create a Visual Story - Visual stories condense complex information into a clear, visual narrative that is easy to grasp. They should highlight the key message and be self-explanatory, minimizing the cognitive effort required to understand. As Alberto Cairo (2012) notes, effective visualizations align data representation with storytelling principles to captivate and inform. Key Question: *Does the visual make a clear point that would take many words to convey?*

Make It a Prop - A visualization becomes a prop when it facilitates engagement and drives meaningful conversations. It should encourage interaction, allowing stakeholders to explore insights collaboratively. Hans Rosling (2006) demonstrated this by using vivid graphics in presentations, turning static data into interactive and memorable experiences. Key Question: *Does the visual serve as a central tool for discussion and insight?*

Emulate Legendary Storytellers - Great visualizations emulate the techniques of master storytellers by being structured, engaging, and emotionally resonant. They guide the audience through a compelling narrative arc, ensuring clarity and impact. Edward Tufte (1983) emphasizes the importance of precision and aesthetics in visual storytelling to enhance understanding. Key Question: *Does the visual embody timeless principles of great storytelling?*

Signs

Jacques Bertin (2011) explains in *Semiology of Graphics* that: “Data is transformed into graphics to understand. A map, a diagram are documents to be interrogated. But understanding means integrating all of the data. In order to do this it's necessary to reduce it to a small number of elementary data. This is the objective of the 'data treatment' be it graphic or mathematic.” We must become good sign makers and send strong unmistakable signals to our audience of the direction they should take in their decision-making.

Semiotics, Signs, and Signifiers - Semiotics, the study of signs, ensures that visuals use symbols and signs appropriately to convey meaning effectively. Signs must align with their intended meaning (signifier) and audience expectations. Jacques Bertin emphasizes the importance of avoiding cultural missteps or confusion to ensure clarity in data visuals (Bertin, 2011). Key Question: *Are signs and symbols used properly in the visual?*

Visualization as a Communication System - A good visualization acts as a communication system, where the message is encoded clearly, transmitted effectively, and decoded without confusion. The signal-to-noise ratio must be high to reduce distractions and ensure the audience grasps the intended meaning (Shron, 2014). Key Question: *Does the visual send a strong unmistakable signal?*

The Functional Art - Art vs. Clarity - Visuals should prioritize functionality over aesthetics. Clarity should never be sacrificed for artistic appeal. Alberto Cairo emphasizes that data visuals are tools to inform and support decisions, not artworks designed to entertain or evoke emotions (Cairo, 2012). Key Question: *Is the chart functionally informational rather than beautiful art?*

Purpose

Stephen Kosslyn (2006) writes in *Graph Design for the Eye and Mind*: “The usefulness of a graph can be evaluated only in the context of the type of data, the questions the designer wants the readers to answer, and the nature of the audience.” We must keep in mind who our audience is, what they want and need to know, and what key-framed analytical questions we answer to inform them.

Information Need - A visualization must directly address the information needs of both the requester and the organization. It should provide data critical to the mission, enabling decision-making and bridging knowledge gaps. Max Shron's framework emphasizes aligning visuals with real organizational problems to ensure decisions are informed and needs are met (Shron, 2014). Key Question: *Does the chart fulfill organizational information needs?*

Audience - Account for the audience's perspective, ensuring clarity and resonance. By doing so, the visual aids the audience in achieving their objectives without confusion or mismatch (Cairo, 2012). Key Question: *Does the chart allow for audience biases, needs, and journey?*

Answer Well-Framed Analytical Questions - Visuals should present key evidence answering well-framed analytical questions developed during analysis. These answers must be clear, actionable, and evident, ensuring the audience grasps the conclusions and their implications for action (Shron, 2014). Key Question: *Does the visual answer a well-framed analytical question?*

Perception

Henry David Thoreau (1884), in his *Journal*, observes that “Many an object is not seen, though it falls within the range of our visual ray, because it does not come within the range of our intellectual ray, i.e. we are not looking for it. So, in the largest sense, we find only the world we look for.” And his key admonition to us as data visualizers is: “The question is not what you look at, but what you see.” Looking happens in the eye, and seeing happens in the mind. *What are we helping our audience to see?*

Use the Eye-Brain System of Seeing - Visuals should guide the viewer's eye to the most critical information unconsciously. Techniques such as Gestalt principles, color theory, and the removal of chart junk can direct attention effectively. Alberto Cairo and Yarbus emphasize designing visuals to leverage the natural shortcuts of the human eye-brain system (Cairo, 2012; Yarbus, 1967).

Key Question: *Does the eye of the viewer focus on the most important point being made?*

Employ the Gestalt Principles of Perception - Gestalt principles ensure the visual design emphasizes figure/ground differences, grouping, connectedness, and flow. These principles guide viewers intuitively to the chart's main message, enhancing clarity and understanding. Effective use of Gestalt psychology helps create visuals that are aesthetically pleasing and informative (Cairo, 2012). Key Question: *Have the principles of the Gestalt psychology of perception been thoughtfully employed in the visual?*

Design with Quality - A quality visual relieves the tension of ignorance by effectively informing the audience. Christopher Alexander's "quality without a name" concept highlights the importance of creating "alive" visuals that satisfy the viewer's need for knowledge through effective design and pattern language (Alexander, 1979). Key Question: *Does the visual resolve the tension in the viewer between not knowing and knowing?*

Method

We must use color appropriately, make sure our charts are devoid of all visual noise, and make sure the PowerPoint slide has a title that gets to the point of your message.

Use Color Effectively - Color enhances visuals when used strategically. Sparing use, semantic correctness, and accessibility for color-blind viewers are essential. Key elements should use color to draw attention while avoiding overuse. Proper application ensures the viewer focuses on critical data without confusion or distraction (Tufte, 1983). Key Question: *Is color used judiciously and sparsely?*

Remove All Chart Junk - Removing unnecessary elements, or "chart junk," sharpens focus and enhances comprehension. Visuals should balance the data-to-ink ratio and eliminate clutter like extraneous grid lines, labels, or colors. Richard Tufte emphasizes simplicity and clarity, ensuring every element contributes to the message (Tufte, 1983). Key Question: *Is the visual clear of unnecessary visual elements not leading to a clear point being made?*

Tell the Story with the Title - A clear, explicit title ensures viewers understand the chart's purpose immediately. Titles should succinctly express the key business point while avoiding ambiguity. McKinsey's method emphasizes direct titling as a storytelling tool to avoid misinterpretation and confusion (McKinsey & Company). Key Question: *Does the title of the chart convey the point being made with the chart?*

Charts

The Charts dimension emphasizes the importance of selecting appropriate chart types and table designs to enhance understanding. By leveraging Cleveland and McGill's functional scale (1987), chart types can be matched to the level of accuracy required for judgment. The chosen chart must effectively answer the business question being posed. Tables, when used, should be readable and enhanced with elements like conditional formatting and Sparklines for emphasis. Visual clarity, functionality, and appropriateness are the cornerstones of effective charting in data visualization.

Use the Right Chart - Cleveland and McGill's functional scale guides the selection of charts by matching the level of encoding to the accuracy required for comparisons. The right chart should align with the purpose of the visualization, the level of judgment needed, and the type of

question being answered (Cleveland & McGill, 1987). Key Question: *Does the type of chart being used match the level of judgment required?*

Select the Chart Type Effectively - Different chart types are suited for specific purposes, such as Pareto charts for contributions or pie charts for parts of a whole. The selected chart must address the intended business question and communicate insights clearly and effectively (Nussbaumer-Knaflic, 2015). Key Question: *Does the chart type used match the business question being answered?*

Enhance Table Data for Emphasis - Tables should prioritize readability and emphasis through effective use of white space, shading, and conditional formatting. Visual enhancements like Sparklines add clarity and insight, transforming tables into powerful tools for communication (Nussbaumer-Knaflic, 2015). Key Question: *Are referenceable visuals (tables) readable with appropriate conditional formatting and thumbnail graphs used for emphasis?*

Dimension	Element	Key Question
Story	Create a Visual Story	Does the visual make a clear point that would take many words to convey?
	Make It a Prop	Does the visual serve as a central tool for discussion and insight?
	Emulate Legendary Storytellers	Does the visual embody timeless principles of great storytelling?
Signs	Bertain: Semiotics, Signs, and Signifiers	Are signs and symbols used properly in the visual?
	Visualization as a Communication System	Does the visual send a strong unmistakable signal?
	Cairo: The Functional Art - Art vs. Clarity	Is the chart functionally informational rather than beautiful art?
Purpose	Information Need	Does the chart fulfill organizational information needs?
	Audience	Does the chart allow for audience biases, needs, and journey?
	Answer Well-Framed Analytical Questions	Does the visual answer a well-framed analytical question?
Perception	Use the Eye-Brain System of Seeing	Does the eye of the viewer focus on the most important point being made?
	Gestalt Principles of Perception	Have the principles of the Gestalt psychology of perception been thoughtfully employed in the visual?
	Design with Quality	Does the visual resolve the tension in the viewer between not knowing and knowing?
Method	Use Color Effectively	Is color used judiciously and sparsely?
	Remove All Chart Junk	Is the visual clear of unnecessary visual elements not leading to a clear point being made?
	Tell the Story with the Title	Does the title of the chart convey the point being made with the chart?
Charts	Right Chart	Does the type of chart being used match the level of judgment required?
	Selection – The Basic Charts	Does the chart type used match the business question being answered?
	Tables - Analyze or Tell?	Are referenceable visuals (tables) readable with appropriate conditional formatting and thumbnail graphs used for emphasis?

Figure 1 – Summary of the six dimensions and 18 elements with the key question to be answered by the data visualizer.

METHOD

This research explored the use of Large Language Models (LLMs) for data visualization analysis using a structured framework. The process consisted of four key phases:

Phase 1 - Framework Training: An LLM (ChatGPT) was fine-tuned using the Fortino Framework document, which outlines 6 dimensions and 18 points for evaluating data visualizations. The framework emphasizes story, signs, purpose, perception, method, and charts as key analytical dimensions.

Phase 2 - LLM Analysis: The trained LLM analyzed five different data visualization cases using the framework. For each case, the LLM provided detailed analysis across all 18 points, including strengths, shortcomings, and suggested improvements. The LLM maintained consistent structure in its analysis, evaluating each dimension systematically.

Phase 3 - SME Validation: A Subject Matter Expert (SME) independently analyzed the same five visualizations using identical framework parameters. The SME analysis served as a benchmark for evaluating the LLM's performance. The SME followed the same structured approach but drew from deep domain expertise in data visualization.

Phase 4 - Comparative Analysis: A separate LLM (Claude) performed detailed comparative analysis between the ChatGPT and SME evaluations. This analysis quantified agreement levels for each framework point using percentage concordance and identified specific areas of alignment and divergence. The comparison revealed overall agreement ranging from 81.4% to 93.3%, with the strongest alignment in functional aspects and the greatest divergence in technical recommendations.

This methodology demonstrated the potential for LLMs to perform structured visualization analysis while highlighting areas where human expertise remains valuable. The high concordance levels suggest LLMs can effectively apply established frameworks, though they may emphasize different aspects than human experts.

The Prompt

You are a data visualization expert tasked with analyzing and improving charts using the principles outlined in the framework found in the "Data Visualization for Business Decisions" book I just uploaded. Your goal is to provide a comprehensive analysis of a given chart across all 6 dimensions and 18 analysis elements, each with specific questions to consider. Here's your process:

- 1. Review the principles in the framework thoroughly, focusing on the 18 dimensions and their associated questions.*
- 2. When presented with a chart, carefully examine it in the context of each dimension.*
- 3. For each dimension: a. Answer all relevant questions from the checklist. b. Identify what aspects of the chart work well in this dimension. c. Pinpoint areas where the chart falls short in this dimension. d. Propose specific improvements to address these shortcomings.*
- 4. After analyzing all 6 dimensions and 18 elements, provide a summary of the key improvements needed, organized by dimension and element.*
- 5. Remember to focus solely on the principles and techniques outlined in the framework of the book, avoiding reference to external sources or methodologies.*
- 6. Maintain a critical yet constructive tone throughout your analysis.*

7. Be specific and detailed in your suggestions for improvement, providing clear rationale based on the book's principles.

8. Consider the chart in the context of its intended purpose and audience, as outlined in the framework.

Are you ready to analyze a chart and give me your opinion?

The Case

The case study is based on set of slides posted by the White House in the United States promoting a certain policy initiative. Early in President Obama's administration there was a push to have all states in the union as well as the Federal Government adopt a higher minimum wage level than was prevalent at that time. One of the agencies of the executive branch is the Council of Economic Advisors (2014). They laid out the economic case for raising the minimum wage. This slide set was made available at the time via the White House website to educate the public, policy makers, and lawmakers. We want to see if we can improve the charts in the presentation which was the chosen communication platform.

Think of yourself as a staff member of the Council of Economic Advisors, the group that created this presentation. One day you are called into the office of Chief Economist and he (or she) says: "Since you are one of our best data visualization experts on staff I would like to look over this document and tell us how we can improve these charts. We want to have the greatest impact for our readers." That's the basis of our analysis.

In this study, we shall undertake the analysis of additional slides from the case study. We have covered two of them in the lab book (slides 2 and 4 from the case study slide set). You can use the four additional examples here to further develop your understanding of how to improve data visual in a business context. To show you the usefulness of the analysis template and the process we will use it as the basis for analysis. You can use the template for yourself with each slide and then test your understanding against the remarks by the expert. Take this an assignment: "If I were given these slides to improve, what would I do along the lines of the six dimensions in the lab book, what would I change?" This way, you are further exercising your newly acquired visualization muscles to create compelling visuals.

The premise for our changes will be that there's a desire to reach and convince a more significant portion of the general population about the benefits of the proposed policy change. As it is, the case study charts were probably created to convince economists and policymakers and other trained people of the proposed policy changes. Although some in the general public could understand the charts, they are perhaps out of the reach of the average citizen. That's where you come in. There is a desire by management to create a new set based on these slides that are more accessible to people who not highly educated. Assume that the new slide set will be designed to be used in a presentation, not as a standalone document. Thus, they are to be a set of props for a presenter rather than a report to be read. Your mission: modify the current slides to fit the new audience and the new purpose.

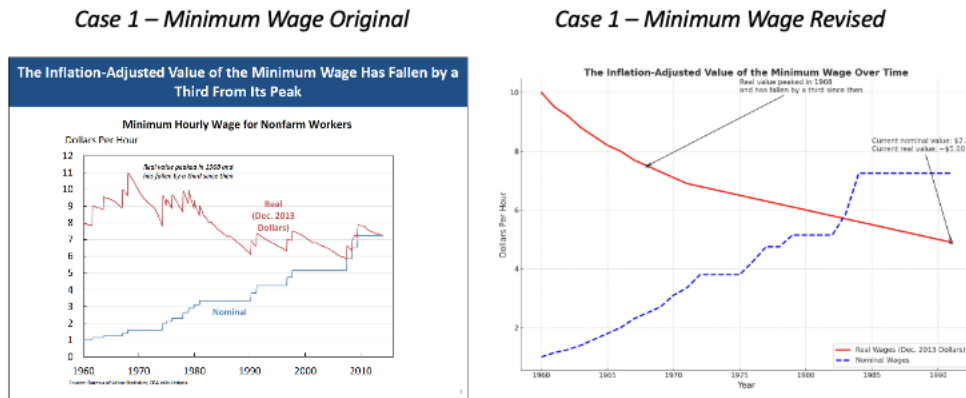


Figure 2 – Case 1 – Minimum Wage. Original chart and ChatGPT revised chart from the White House Economic Advisors Case dataset.

Example Outcome from ChatGPT

The analysis of Case 1 (minimum wage visualization) demonstrates ChatGPT's systematic application of the 6-dimension framework. ChatGPT's detailed assessment identified key strengths (clear trend visualization, appropriate chart type) and weaknesses (poor annotation visibility, unclear terminology for general audiences). The chatbot provided specific, actionable improvements focusing on accessibility and clarity, such as simplifying the title, enhancing line visibility, and adding explanatory notes for terms like "real" versus "nominal" wages. The revised visualization (Figure 2) shows these implemented changes, featuring clearer annotations, stronger color contrast, and simplified messaging. This analysis showcases both ChatGPT's ability to systematically evaluate visualizations and its practical capacity to generate improved versions, though some technical nuances noted by the SME were missed in the AI analysis.

ChatGPT's output advice to improve the visual:

"Story

"Visual Story: Strengths: The chart successfully conveys a clear story about the decline in the real value of the minimum wage since its peak in 1968. Shortcomings: The story could be stronger if the decline's implications for workers were explicitly linked to real-world outcomes. Improvements: Add a brief annotation or subtitle explaining the impact of the declining real value on workers' purchasing power."

"Visual Props: Strengths: The chart uses lines to compare real and nominal wages effectively. Shortcomings: Text annotations on the chart are somewhat small and may not grab the viewer's attention immediately. Improvements: Enlarge and bold the annotations to make them more visible and emphasize the key message."

"Storytellers: Strengths: The chart aligns with historical data storytelling techniques (e.g., timelines and comparisons). Shortcomings: Does not directly emulate iconic visuals for economic trends (e.g., Hans Rosling's engaging style or Florence Nightingale's clarity)."

Improvements: Incorporate elements such as callouts or a more engaging color scheme to make the visual more dynamic.

“Signs

“Signs: Strengths: Symbols (lines and labels) are used effectively to distinguish real and nominal values. Shortcomings: The red line for real wages is clear, but the blue line for nominal wages blends into the background. Improvements: Use a darker shade of blue or add a stronger outline to improve visibility.

Communication: Strengths: The signal-to-noise ratio is reasonably high as the chart focuses on its central message. Shortcomings: The title is slightly long and could distract from the chart. Improvements: Simplify the title to something like "Real Minimum Wage Has Dropped by One-Third Since 1968."

Function: Strengths: The chart prioritizes clarity over aesthetic appeal, which is appropriate for the audience. Shortcomings: The design is functional but could be more visually engaging without sacrificing clarity. Shortcomings: Add subtle visual effects, such as shading or arrows, to emphasize key points.

“Purpose

“Need - Strengths: The chart fulfills its purpose of showing the historical decline in real wages. Shortcomings: It doesn't clearly address why this decline is important or what action should follow. Improvements: Add a concluding sentence or callout to connect the data to the advocacy for raising the minimum wage.

“Audience - Strengths: The chart is designed for policymakers and an informed audience. Shortcomings: The general public may struggle with interpreting "real" versus "nominal" values without a brief explanation. Improvements: Include a simple legend or brief note defining "real" and "nominal."

Frame - Strengths: The chart answers a well-framed analytical question about the historical trend of minimum wages. Shortcomings: The framing could be stronger if it directly tied the data to actionable insights. Improvements: Frame the title or subtitle as a question, e.g., "Why Has the Real Minimum Wage Declined by One-Third Since 1968?"

“Perception

“Seeing - Strengths: The red line for real wages naturally draws the eye, emphasizing the most critical trend. Shortcomings: The gridlines and axis labels are visually prominent, potentially distracting from the data. Improvements: Reduce the intensity of gridlines and make axis labels more subtle.

“Mind - Strengths: Gestalt principles, such as continuity and grouping, are evident in the use of lines. Shortcomings: The figure-ground distinction could be improved; the chart lacks a clear emphasis on the starting and ending points of the trend. Improvements: Highlight the peak and trough points with markers or annotations to improve focus.

“Quality - Strengths: The chart informs the viewer and dispels some ignorance about historical wage trends. Shortcomings: It doesn’t fully resolve the tension of understanding how these trends affect workers today. Improvements: Include a brief note or callout explaining the current purchasing power of minimum wages in 2013 dollars.

“Method

“Color - Strengths: Red and blue are used to differentiate real and nominal wages. Shortcomings: The blue line’s lighter color makes it less impactful. Improvements: Use a darker blue or bold the line for nominal wages.

“Chart Junk - Strengths: The chart avoids unnecessary elements like decorative icons or excessive annotations. Shortcomings: The text annotations could be perceived as clutter because they are small and overlap the lines. Improvements: Relocate annotations to the margin or use callout boxes.

“Title - Strengths: The title captures the essence of the chart’s message. Shortcomings: It’s long and somewhat redundant given the data labels on the chart. Improvements: Simplify the title to be more direct and impactful.

“Charts

“Right Chart - Strengths: The line chart is an appropriate choice for showing trends over time. Shortcomings: It might not convey the dramatic nature of the decline as effectively as an area or bar chart could. Improvements: Consider an area chart to emphasize the magnitude of the decline in real wages.

“Selection - Strengths: The chart type matches the question of wage trends over time. Shortcomings: It could better highlight key milestones, such as policy changes affecting wages. Improvements: Add vertical markers or shaded regions to indicate key historical events or policies.

“Tables - Strengths: Not applicable here as there are no tables in the visual. Shortcomings: None.

“Summary of Key Improvements

“Title: Simplify to emphasize the key message more directly.

“Annotations: Enlarge or relocate text annotations for better visibility and focus.

“Color: Adjust the blue line to make it more distinct.

“Emphasis: Highlight key points on the lines (e.g., peaks and troughs) with markers or arrows.

“Audience Clarity: Add definitions or notes for “real” and “nominal” to aid comprehension.

“Context: Connect the trends to present-day implications or actions.”

RESULTS

The comparison between ChatGPT and SME analyses using the Fortino Framework reveals strong overall alignment, with concurrence ranging from 81.4% to 93.3%. The analyses show the strongest agreement in evaluating fundamental visualization principles, particularly in

function over decoration, basic chart type selection, and table analysis considerations. The SME consistently provides more historically grounded and technical recommendations, often referencing visualization pioneers like Playfair and emphasizing consistency with existing presentation sets. In contrast, ChatGPT tends to focus on modern design principles, accessibility considerations, and alternative visualization methods.

Five charts were selected from the White Hour Advisor's case that are representative of the most basic data visualization types: line, bar, pie, and time series. The first chart can be seen in Figure 1. The other four charts from the case are shown in Figure 2.

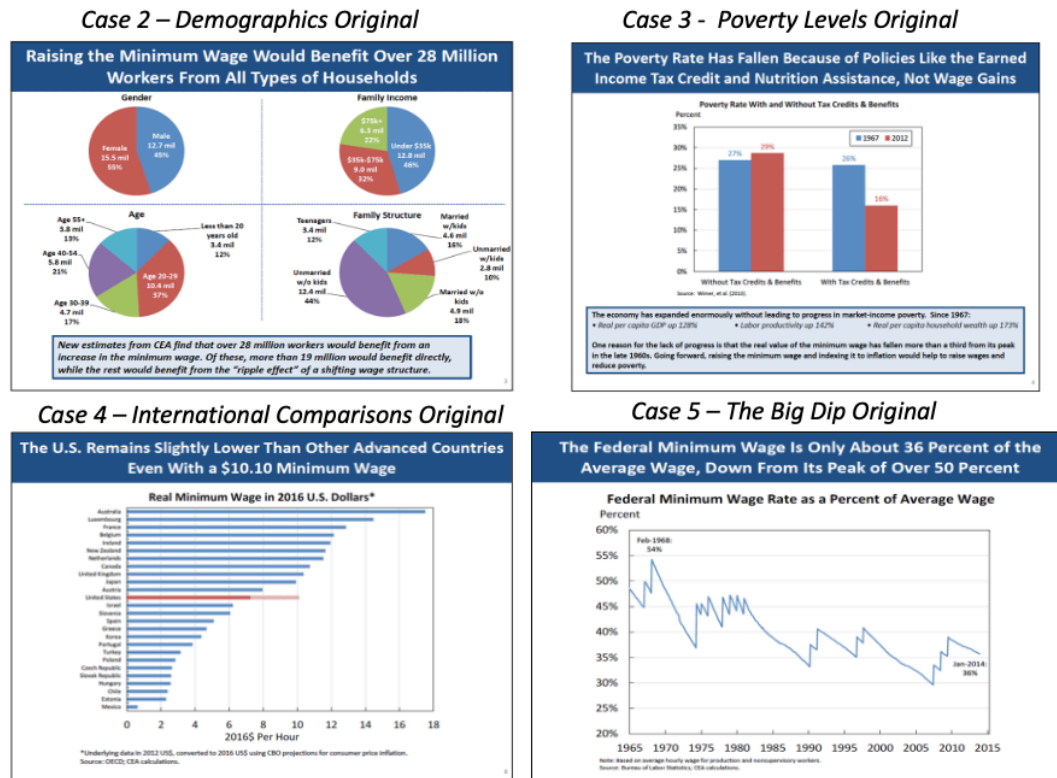


Figure 3 – Original Charts from the White House Economic Advisors Case dataset.

Key disagreements emerge in complex chart type recommendations, where the SME often favors traditional approaches while ChatGPT suggests modern alternatives. Notable differences also appear in color scheme recommendations, with ChatGPT emphasizing accessibility for colorblind viewers while the SME focuses on maintaining existing presentation schemes. The highest concurrence (93.3%) appears in Case 4's international comparison analysis, where both assessors strongly agreed on the effectiveness of the visualization, while the lowest agreement (81.4%) occurs in Case 2's demographic analysis, primarily due to divergent views on pie chart usage versus alternative formats.

Case Study	Overall Agreement	Strongest Agreement Areas	Major Disagreements
Case 1 (Minimum Wage)	85.5%	Function, Frame, Title	Chart type recommendations, Technical depth
Case 2 (Demographics)	81.4%	Function, Tables, Purpose	Chart type (pie vs. bar), Color usage
Case 3 (Poverty Levels)	84.2%	Function, Tables, Title	Chart type (line vs. bar), Visual emphasis
Case 4 (International Comparison)	93.3%	Function, Tables, Story	Color accessibility, Data grouping
Case 5 (Big Dip)	88.3%	Function, Tables, Communication	Historical context, Technical specificity

Figure 4 – Results of comparing SME analysis to ChatGPT analysis based on the 6-dimension framework for five separate visuals from the case.

The resulting reworked charts by ChatGPT are shown in Figure 5 for Cases 2 to 4. The reworked chart result for Case 1 is displayed in Figure 1. These reworked charts all show several characteristics: (1) they were generated by ChatGPT using Python code, (2) they appear to be an improvement on the original and useful for fulfilling the request to make them more understandable to the general public, and (3) they could use additional work to enhance them more compelling.

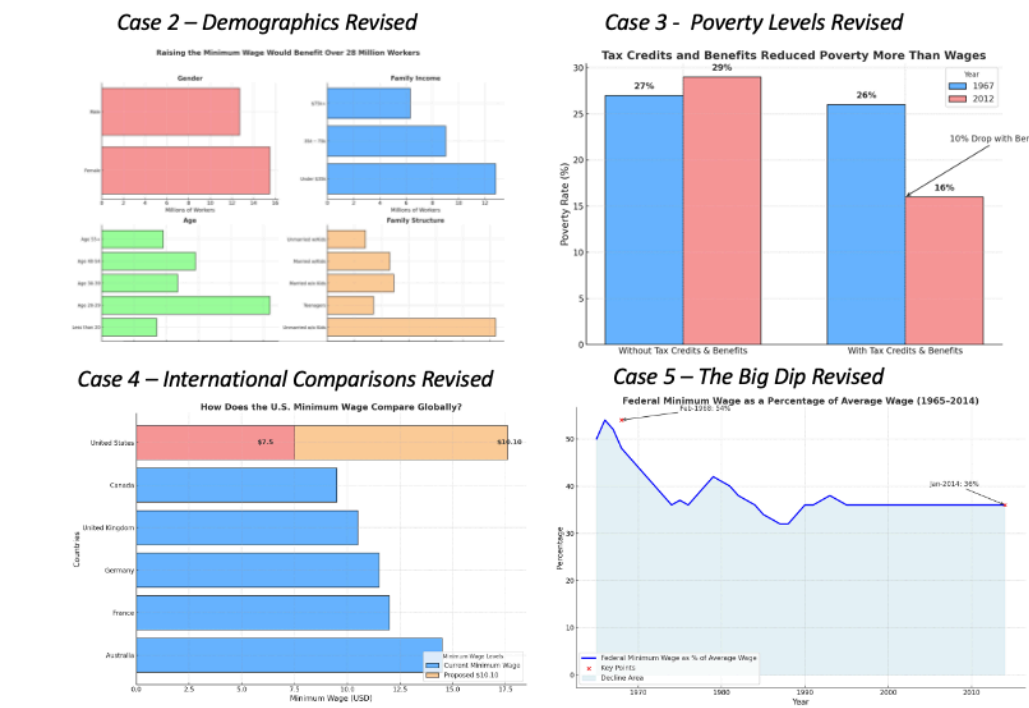


Figure 5 – ChatGPT revised charts from the White House Economic Advisors Case dataset.

DISCUSSION

The results of this exploratory study suggest that an AI-driven chatbot can meaningfully contribute to data visualization analysis and improvement, though with important caveats and limitations. The relatively high agreement rates (81-93%) between chatbot and expert analyses indicate promising potential, particularly in basic functional assessments and accessibility improvements. However, these findings should be interpreted as preliminary evidence rather than definitive proof of effectiveness. The chatbot's occasional divergence from expert recommendations, especially in technical aspects and traditional visualization approaches, highlights areas needing refinement. While the improved charts demonstrate enhanced clarity for general audiences, this proof-of-concept study primarily establishes the feasibility of using AI as a visualization coach rather than validating it as a replacement for human expertise. Further research is needed to rigorously validate these initial findings, particularly through controlled studies with larger samples and diverse visualization types. The results suggest this approach merits additional investigation while acknowledging that the current implementation represents just a first step toward developing reliable AI-assisted visualization tools.

CONCLUSIONS AND FURTHER WORK

This exploratory study demonstrated the feasibility of using LLMs as visualization coaches using a structured analytical framework. Our initial results, showing 81-93% agreement with expert analysis, suggest that AI can meaningfully assist in improving data visualizations while highlighting areas needing refinement.

Future work will focus on (1) Evaluating alternative frameworks, including Stephen Few's methodology (2012) and ChatGPT's native visualization knowledge; (2) Implementing Batova's 5-second and task-based testing protocols for more rigorous validation; (3) Conducting controlled usability studies comparing original and AI-improved visualizations; (4) Expanding the test dataset to include diverse visualization types and complexity levels; and (5) Investigating the role of domain expertise in AI-assisted visualization enhancement.

A key priority will be validating whether AI-improved visualizations consistently enhance understanding and decision-making across different user groups and contexts. This will include measuring both objective performance metrics and subjective user experience factors. We also aim to explore how AI coaching can be effectively integrated into existing visualization workflows while maintaining appropriate human oversight and creative control. This includes developing guidelines for when and how to apply AI suggestions most effectively.

REFERENCES

- Alexander, C. (1979), *The timeless way of building*. Vol. 1. New York: Oxford University Press, 1979.
- Argyris, C., and Schon, D. (1974). *Theory in practice: Increasing professional effectiveness*. Jossey-Bass, 1974.
- Batova, T. (2024). *Usability testing for data visualizations* (UVA-BC-0301). Darden School Foundation, University of Virginia.
- Bertin, J. (2011), "General theory, from semiology of graphics." *The Map Reader: Theories of Mapping Practice and Cartographic Representation* (2011): 8-16.
- Cairo, A. (2012), *The Functional Art: An introduction to information graphics and visualization*. New Riders, 2012.

- Cleveland, W., and McGill, R. (1987), "Graphical perception: The visual decoding of quantitative information on graphical displays of data." *Journal of the Royal Statistical Society. Series A (General)* (1987): 192-229.
- Cotgreave, A. (2013), "The 5 Most Influential Data Visualizations of All Time", Tableau, 2013, <https://www.tableau.com/learn/webinars/5-most-influential-visualizations-all-time>.
- Council of Economic Advisors. (2014). *The economic case for raising the minimum wage*. The White House. <https://www.slideshare.net/slideshow/the-economic-case-for-raising-the-minimum-wage/31128477>
- Few, S. (2009), *Now you see it: simple visualization techniques for quantitative analysis*. Analytics Press, 2009.
- Few, S. (2012). *Show Me the Numbers: Designing Tables and Graphs to Enlighten* (2nd ed.). Analytics Press, 2012.
- Fortino, A. (2020). *Data Visualization for Business Decisions: A Laboratory Manual*. Mercury Publishing, 2020.
- Kosslyn, S. (2006), *Graph design for the eye and mind*. OUP USA, 2006.
- Lewis, R. (2010). *When cultures collide*. Nicholas Brealey Publishing, 2010.
- Medina, J. (2012). *Brain Rules*. Pear Press, Seattle, 2012.
- Nussbaumer Knaflic, C. (2015), "Story Telling with Data: Visualizing Opportunity", in September 16, 2015, Makeovers, <http://www.storytellingwithdata.com/blog/2015/9/16/visualizing-opportunity>.
- Rosling, H. (2006), "Hans Rosling shows the best stats you've ever seen, 2006." *Internet*: http://www.ted.com/talks/hans_rosling_shows_the_best_stats_you_ve_ever_seen.html.
- Shalloway, A. (2011). Visualizations act as a campfire around which we gather to tell stories. *Net Objectives*. Retrieved from <https://visualign.org/quotes/>
- Shron, M. (2014), *Thinking with Data: How to Turn Information into Insights*. O'Reilly Media, Inc., 2014.
- Thoreau, H. (1884), *The Writings of Henry David Thoreau (Volume V)*. (1884).
- Tufte, E., and Graves-Morris, P. (2014), "The visual display of quantitative information; 1983." (2014).
- Yarbus, A. (1967), "Eye movements during perception of complex objects." *Eye movements and vision*, pp. 171-211. Springer, Boston, MA, 1967.